

Лабораторная работа

Номер 7

Мошаров Д.М.

01 января 1970

Российский университет дружбы народов, Москва, Россия

Информация

- Мошаров Денис Максимович
- Студент
- Российский университет дружбы народов

Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Создание нового проекта в GNS3

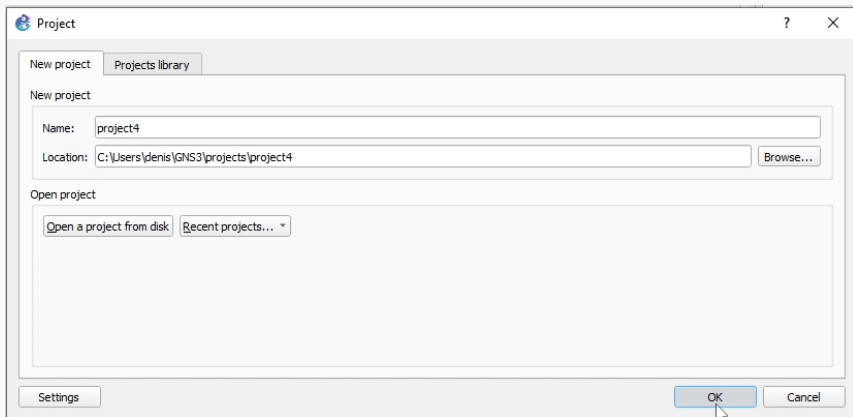


Рис. 1: Создание нового проекта в GNS3

Топология сети для настройки DHCPv4

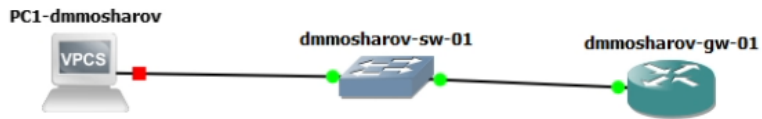


Рис. 2: Топология сети для настройки DHCPv4

Запуск захвата трафика в Wireshark

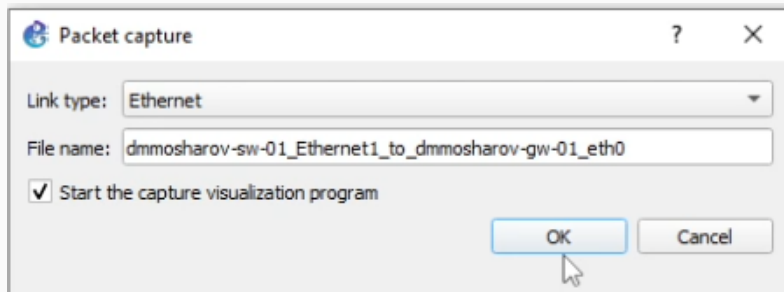
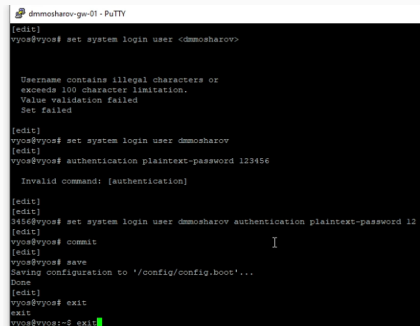


Рис. 3: Запуск захвата трафика в Wireshark

Базовая настройка маршрутизатора VyOS



```
[edit]
vyos@vyos# set system login user <dmmosharov>

Username contains illegal characters or
exceeds 100 character limitation.
Value validation failed
Set failed

[edit]
vyos@vyos# set system login user dmmosharov
[edit]
vyos@vyos# authentication plaintext-password 123456

Invalid command: [authentication]

[edit]
[edit]
3456@vyos# set system login user dmmosharov authentication plaintext-password 12
[edit]
vyos@vyos# commit
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
```

Рис. 4: Базовая настройка маршрутизатора VyOS

Удаление стандартного пользователя

```
dmmosharov@dmmosharov-gw-01:~$ configure
[edit]
dmmosharov@dmmosharov-gw-01# delete system login user vyos
[edit]
dmmosharov@dmmosharov-gw-01# commit
[edit]
dmmosharov@dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dmmosharov@dmmosharov-gw-01#
```

Рис. 5: Удаление стандартного пользователя

Настройка IPv4 адресации и DHCP-сервера

```
dmmosharov@dmmosharov-gw-01# delete system login user vyos
[edit]
dmmosharov@dmmosharov-gw-01# commit
[edit]
dmmosharov@dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dmmosharov@dmmosharov-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
rov domain-name dmmosharov.net set service dhcp-server shared-network-name dmmosha
[edit]
rov name-server 10.0.0.1
[edit]
dmmosharov@dmmosharov-gw-01# set service dh
dhcp-relay      dhcp-server      dhcpv6-relay  dhcpv6-server
[edit]
rov subnet 10.0.0.0/24 default-router 10.0.0.1

Configuration path: service dhcp-server shared-network-name dmmosharov [subnet] is not valid
Set failed

[edit]
rov subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
[edit]
[edit]
rov subnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
rov subnet 10.0.0.0/24 range hosts stop 10.0.0.25 set service dhcp-server shared-network-name dmmosha
```

Рис. 6: Настройка IPv4 адресации и DHCP-сервера

Проверка статистики DHCP-сервера

```
exit
dmmosharov@dmmosharov-gw-01:~$ show dhcp server statistics
Pool          Size    Leases   Available  Usage
-----
dmmosharov    252      0        252      0%
dmmosharov@dmmosharov-gw-01:~$ show dhcp server leases
IP address    Hardware address  State   Lease start   Lease expiration  Remaining  Pool  Hostname
-----
dmmosharov@dmmosharov-gw-01:~$
```

Рис. 7: Проверка статистики DHCP-сервера

Получение IP-адреса на клиенте PC1 (процесс DORA)

```
VPCS> [0] dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = dmcaharov.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = VPCS
```

Рис. 8: Получение IP-адреса на клиенте PC1 (процесс DORA)

Проверка списка выданных адресов (Leases)

```
exit
dmmosharov@dmmosharov-gw-01:~$ show dhcp server statistics
Pool          Size    Leases   Available  Usage
-----
dmmosharov    252     0        252        0%
dmmosharov@dmmosharov-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Hostname
-----
dmmosharov@dmmosharov-gw-01:~$ show dhcp server statistics
Pool          Size    Leases   Available  Usage
-----
dmmosharov    252     1        251        0%
dmmosharov@dmmosharov-gw-01:~$ show dhcp server leases
```

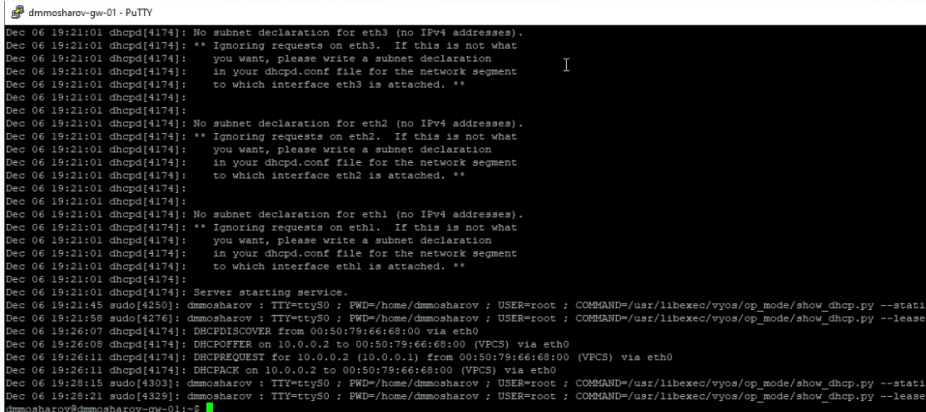
Рис. 9: Проверка списка выданных адресов (Leases)

Просмотр логов DHCP-сервера

```
nsandryushin@nsandryushin-gw-01:~$ show dhcp server statistics
Pool      Size      Leases      Available  Usage
-----
nsandryushin 252        1            251      0%
nsandryushin@nsandryushin-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration
Remaining  Pool              Hostname
-----
10.0.0.2    00:50:79:66:68:00 active  2025/12/06 13:37:06 2025/12/07 13:37:06
23:58:40    nsandryushin VPCS
nsandryushin@nsandryushin-gw-01:~$ show log | grep dhcp
Dec 06 13:10:46 dhclient-script-vyos[1615]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:47 vyos-hostsd[607]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 06 13:10:47 dhclient-script-vyos[1615]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:48 vyos-hostsd[607]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 06 13:10:53 dhclient-script-vyos[1776]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:55 vyos-hostsd[607]: Request data: {"type": "search_domains", "op": "delete",
```

Рис. 10: Просмотр логов DHCP-сервера

Детальный анализ логов работы DHCP



```
dmmosharov-gw-01 - PuTTY
Dec 06 19:21:01 dhcpd[4174]: No subnet declaration for eth3 (no IPv4 addresses).
Dec 06 19:21:01 dhcpd[4174]: ** Ignoring requests on eth3. If this is not what
Dec 06 19:21:01 dhcpd[4174]: you want, please write a subnet declaration
Dec 06 19:21:01 dhcpd[4174]: in your dhcpd.conf file for the network segment
Dec 06 19:21:01 dhcpd[4174]: to which interface eth3 is attached. **
Dec 06 19:21:01 dhcpd[4174]:
Dec 06 19:21:01 dhcpd[4174]: No subnet declaration for eth2 (no IPv4 addresses).
Dec 06 19:21:01 dhcpd[4174]: ** Ignoring requests on eth2. If this is not what
Dec 06 19:21:01 dhcpd[4174]: you want, please write a subnet declaration
Dec 06 19:21:01 dhcpd[4174]: in your dhcpd.conf file for the network segment
Dec 06 19:21:01 dhcpd[4174]: to which interface eth2 is attached. **
Dec 06 19:21:01 dhcpd[4174]:
Dec 06 19:21:01 dhcpd[4174]: No subnet declaration for eth1 (no IPv4 addresses).
Dec 06 19:21:01 dhcpd[4174]: ** Ignoring requests on eth1. If this is not what
Dec 06 19:21:01 dhcpd[4174]: you want, please write a subnet declaration
Dec 06 19:21:01 dhcpd[4174]: in your dhcpd.conf file for the network segment
Dec 06 19:21:01 dhcpd[4174]: to which interface eth1 is attached. **
Dec 06 19:21:01 dhcpd[4174]:
Dec 06 19:21:01 dhcpd[4174]: Server starting service.
Dec 06 19:21:45 sudo[4250]: dmmosharov : TTY=ttys0 ; FWD=/home/dmmosharov ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --stat
Dec 06 19:21:58 sudo[4276]: dmmosharov : TTY=ttys0 ; FWD=/home/dmmosharov ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --lease
Dec 06 19:26:07 dhcpd[4174]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Dec 06 19:26:08 dhcpd[4174]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Dec 06 19:26:11 dhcpd[4174]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (VPCS) via eth0
Dec 06 19:26:11 dhcpd[4174]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0
Dec 06 19:28:15 sudo[4303]: dmmosharov : TTY=ttys0 ; FWD=/home/dmmosharov ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --stat
Dec 06 19:28:21 sudo[4329]: dmmosharov : TTY=ttys0 ; FWD=/home/dmmosharov ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --lease
dmmosharov@dmmosharov-gw-01:~$
```

Рис. 11: Детальный анализ логов работы DHCP

Анализ DHCP пакетов в Wireshark

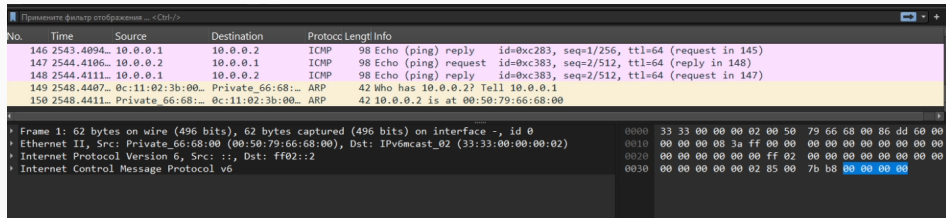


Рис. 12: Анализ DHCP пакетов в Wireshark

Расширенная топология для настройки DHCPv6

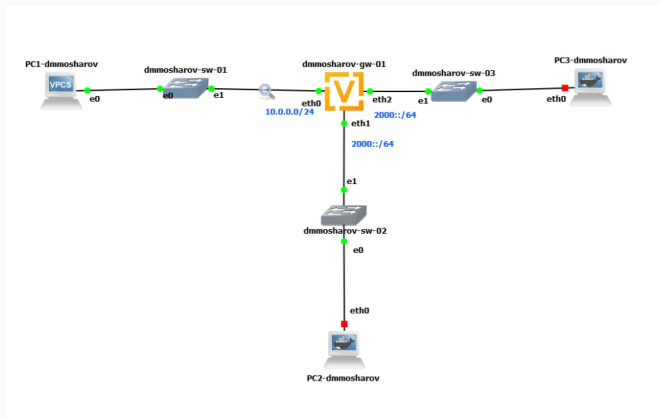


Рис. 13: Расширенная топология для настройки DHCPv6

Настройка IPv6 адресов на интерфейсах

```
[edit]
dmmosharov@dmmosharov-gw-01# set interfaces ethernet eth1

Configuration path: [interfaces ethernet eth1] already exists

[edit]
dmmosharov@dmmosharov-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
dmmosharov@dmmosharov-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
dmmosharov@dmmosharov-gw-01# show interface
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:11:02:3b:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:11:02:3b:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:11:02:3b:00:02
  }
  ethernet eth3 {
```

Рис. 14: Настройка IPv6 адресов на интерфейсах

Настройка Stateless DHCPv6 на интерфейсе eth1

```
[edit]
dmmosharov@dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
0::/64arov@dmmosharov-gw-01# set service router-advert interface eth1 prefix 200
[edit]
[edit]
ig-flagrov@dmmosharov-gw-01# set service router-advert interface eth1 other-conf
[edit]
harov-stateless subnet 2000::0/64service dhcpv6-server shared-network-name dmmos
[edit]
[edit]
harov-stateless common-options name-server 2000::1rver shared-network-name dmmos
[edit]
harov-stateless common-options domain-search dmmosharov.neted-network-name dmmos
[edit]
dmmosharov@dmmosharov-gw-01# commit
[edit]
dmmosharov@dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dmmosharov@dmmosharov-gw-01#
```

Рис. 15: Настройка Stateless DHCPv6 на интерфейсе eth1

Просмотр конфигурации маршрутизатора

```
[edit]
dmmosharov@dmmosharov-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:11:02:3b:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:11:02:3b:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:11:02:3b:00:02
    }
    ethernet eth3 {
        hw-id 0c:11:02:3b:00:03
    }
    ethernet eth4 {
        hw-id 0c:11:02:3b:00:04
    }
    ethernet eth5 {
        hw-id 0c:11:02:3b:00:05
    }
    ethernet eth6 {
```

Рис. 16: Просмотр конфигурации маршрутизатора

Проверка автоконфигурации IPv6 (SLAAC) на PC2

```
root@PC2-dmmosharov: ~  
fe80::/64 :: U 256 1 0 eth0  
fe80::/64 :: U 256 1 0 eth1  
::/0 fe80::e11:2ff:fe3b:1 UGD Ae 1024 1 0 et  
h0  
::1/128 :: Un 0 3 0 lo  
2000::42:e2ff:fe06:8300/128 :: Un 0 2 0 eth0  
fe80::42:e2ff:fe06:8300/128 :: Un 0 3 0 eth0  
fe80::42:e2ff:fe06:8301/128 :: Un 0 2 0 eth1  
ff00::/8 :: U 256 3 0 eth0  
ff00::/8 :: U 256 1 0 eth1  
::/0 :: !n -1 1 0 lo  
  
(root@ PC2-dmmosharov)~  
# ping 2000::1 -c 2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=1.16 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=0.643 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 100lms  
rtt min/avg/max/mdev = 0.643/0.900/1.158/0.257 ms  
  
(root@ PC2-dmmosharov)~  
#
```

Рис. 17: Проверка автоконфигурации IPv6 (SLAAC) на PC2

Получение DNS-настроек через Stateless DHCPv6 на PC2

```
root@PC2-dmmosharov: ~  
Created duid "\000\003\000\001\002B\342\006\203\000".  
PRC: Requesting information (INIT).  
XMT: Forming Info-Request, 0 ms elapsed.  
XMT: Info-Request on eth0, interval 930ms.  
RCV: Reply message on eth0 from fe80::ell:2ff:fe3b:1.  
PRC: Done.  
  
(root@PC2-dmmosharov)-[~]  
# ping 2000::1 -c 2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=0.688 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=0.516 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1020ms  
rtt min/avg/max/mdev = 0.516/0.602/0.688/0.086 ms  
  
(root@PC2-dmmosharov)-[~]  
# cat /etc/resolv.conf  
search dmmosharov.net.  
nameserver 2000::1  
  
(root@PC2-dmmosharov)-[~]  
#
```

Рис. 18: Получение DNS-настроек через Stateless DHCPv6 на PC2

Проверка таблицы аренды DHCPv6 (Stateless)

```
[edit]
dmmosharov@dmmosharov-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication  Lease expiration  Remaining  Type  Pool  IAID_DUID
-----
[edit]
dmmosharov@dmmosharov-gw-01#
```

Рис. 19: Проверка таблицы аренды DHCPv6 (Stateless)

Анализ пакетов Stateless DHCPv6 в Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
27	447.165852	2000::42:e2ff:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 28)
28	447.166314	2000::1	2000::42:e2ff:...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=2, hop limit=64 (request in 27)
29	451.632782	fe80::e11:2ff:...	2000::42:e2ff:...	ICMPv6	86	Neighbor Solicitation for 2000::42:e2ff:fe06:8300 from 0c:11:02:3b:00:01
30	451.632979	2000::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	78	Neighbor Advertisement 2000::42:e2ff:fe06:8300 (sol)
31	456.795919	fe80::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	86	Neighbor Solicitation for fe80::e11:2ff:fe3b:1 from 02:42:e2:06:83:00
32	456.796548	fe80::e11:2ff:...	fe80::42:e2ff:...	ICMPv6	78	Neighbor Advertisement fe80::e11:2ff:fe3b:1 (rtr, sol)
33	461.873533	fe80::e11:2ff:...	fe80::42:e2ff:...	ICMPv6	86	Neighbor Solicitation for fe80::42:e2ff:fe06:8300 from 0c:11:02:3b:00:01
34	461.873657	fe80::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	78	Neighbor Advertisement fe80::42:e2ff:fe06:8300 (sol)
35	808.496093	fe80::42:e2ff:...	ff02::1:2	DHCPv6	98	Information-request XID: 0x7b23c6 CID: 000300010242e2068300
36	808.497624	fe80::e11:2ff:...	fe80::42:e2ff:...	DHCPv6	138	Reply XID: 0x7b23c6 CID: 000300010242e2068300
37	813.617028	fe80::e11:2ff:...	fe80::42:e2ff:...	ICMPv6	86	Neighbor Solicitation for fe80::42:e2ff:fe06:8300 from 0c:11:02:3b:00:01
38	813.617384	fe80::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	78	Neighbor Advertisement fe80::42:e2ff:fe06:8300 (sol)
39	818.784208	fe80::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	86	Neighbor Solicitation for fe80::e11:2ff:fe3b:1 from 02:42:e2:06:83:00
40	818.784718	fe80::e11:2ff:...	fe80::42:e2ff:...	ICMPv6	78	Neighbor Advertisement fe80::e11:2ff:fe3b:1 (rtr, sol)
41	822.400255	2000::42:e2ff:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0002, seq=1, hop limit=64 (reply in 42)
42	822.400825	2000::1	2000::42:e2ff:...	ICMPv6	118	Echo (ping) reply id=0x0002, seq=1, hop limit=64 (request in 41)
43	823.419809	2000::42:e2ff:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0002, seq=2, hop limit=64 (reply in 44)
44	823.420191	2000::1	2000::42:e2ff:...	ICMPv6	118	Echo (ping) reply id=0x0002, seq=2, hop limit=64 (request in 43)
45	827.441046	fe80::e11:2ff:...	2000::42:e2ff:...	ICMPv6	86	Neighbor Solicitation for 2000::42:e2ff:fe06:8300 from 0c:11:02:3b:00:01
46	827.441144	2000::42:e2ff:...	fe80::e11:2ff:...	ICMPv6	78	Neighbor Advertisement 2000::42:e2ff:fe06:8300 (sol)
47	827.487779	fe80::42:e2ff:...	2000::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 02:42:e2:06:83:00
48	827.488127	2000::1	fe80::42:e2ff:...	ICMPv6	78	Neighbor Advertisement 2000::1 (rtr, sol)
49	836.629978	fe80::e11:2ff:...	ff02::1	ICMPv6	118	Router Advertisement from 0c:11:02:3b:00:01

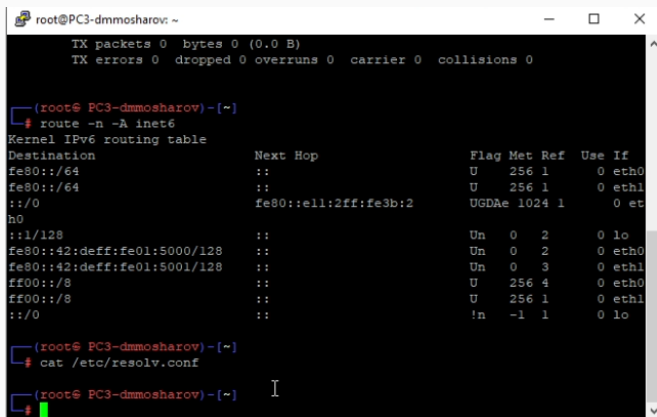
Рис. 20: Анализ пакетов Stateless DHCPv6 в Wireshark

Настройка Stateful DHCPv6 на интерфейсе eth2

```
[edit]
harov-stateful@harov-gw-01# set service dhcpv6-server shared-network-name dmmos
[edit]
harov-stateful subnet 2001::0/64
[edit]
harov-stateful subnet 2001::0/64 name-server 2001::1
[edit]
harov-stateful subnet 2001::0/64 domain-search dmmosharov.net-network-name dmmos
[edit]
harov-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199mos
[edit]
dmmosharov@dmmosharov-gw-01# commit
[edit]
dmmosharov@dmmosharov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
dmmosharov@dmmosharov-gw-01#
```

Рис. 21: Настройка Stateful DHCPv6 на интерфейсе eth2

Начальное состояние сети на PC3

A terminal window titled 'root@PC3-dmmosharov: ~' with standard window controls. It displays network statistics at the top, followed by the command 'route -n -A inet6' and its output, which is the Kernel IPv6 routing table. The table lists destinations, next hops, flags, metrics, references, and interfaces. Finally, the command 'cat /etc/resolv.conf' is entered, and the cursor is positioned at the start of a new line.

```
root@PC3-dmmosharov: ~
TX packets 0  bytes 0 (0.0 B)
TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

(root@ PC3-dmmosharov)-[~]
# route -n -A inet6
Kernel IPv6 routing table
Destination                Next Hop                    Flag Met Ref  Use If
fe80::/64                   ::                          U    256 1    0 eth0
fe80::/64                   ::                          U    256 1    0 eth1
::/0                       fe80::e11:2ff:fe3b:2       UGDAe 1024 1    0 et
h0
::1/128                     ::                          Un    0  2    0 lo
fe80::42:deff:fe01:5000/128 ::                          Un    0  2    0 eth0
fe80::42:deff:fe01:5001/128 ::                          Un    0  3    0 eth1
ff00::/8                   ::                          U    256 4    0 eth0
ff00::/8                   ::                          U    256 1    0 eth1
::/0                       ::                          !n   -1  1    0 lo

(root@ PC3-dmmosharov)-[~]
# cat /etc/resolv.conf

(root@ PC3-dmmosharov)-[~]
#
```

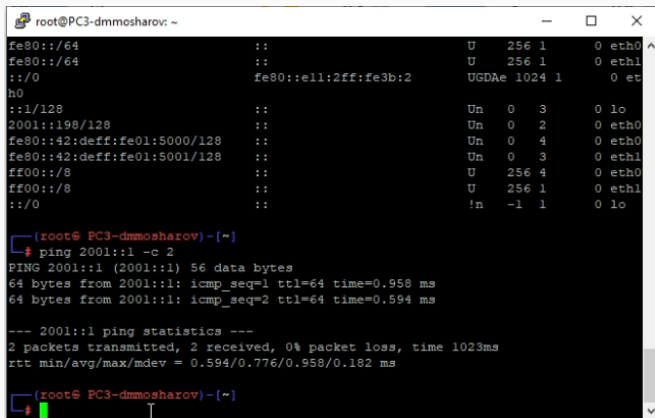
Рис. 23: Начальное состояние сети на PC3

Получение адреса через Stateful DHCPv6 на PC3

```
(root@ PC3-dmmosharov)-[~]  
# dhclient -6 -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2022 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created duid "\000\001\000\0010\307K\000\002B\336\001P\000".  
PRC: Soliciting for leases (INIT)↓
```

Рис. 24: Получение адреса через Stateful DHCPv6 на PC3

Проверка сетевых настроек на PC3 после работы DHCPv6



```
root@PC3-dmmosharov: ~  
fe80::/64 :: U 256 1 0 eth0  
fe80::/64 :: U 256 1 0 eth1  
::/0 fe80::e11:2ff:fe3b:2 UGD Ae 1024 1 0 et  
h0  
::1/128 :: Un 0 3 0 lo  
2001::198/128 :: Un 0 2 0 eth0  
fe80::42:deff:fe01:5000/128 :: Un 0 4 0 eth0  
fe80::42:deff:fe01:5001/128 :: Un 0 3 0 eth1  
ff00::/8 :: U 256 4 0 eth0  
ff00::/8 :: U 256 1 0 eth1  
::/0 :: !n -1 1 0 lo  
  
(root@PC3-dmmosharov)~  
# ping 2001::1 -c 2  
PING 2001::1 (2001::1) 56 data bytes  
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=0.958 ms  
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=0.594 ms  
  
--- 2001::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1023ms  
rtt min/avg/max/mdev = 0.594/0.776/0.958/0.182 ms  
  
(root@PC3-dmmosharov)~  
#
```

Рис. 25: Проверка сетевых настроек на PC3 после работы DHCPv6

Проверка таблицы аренды DHCPv6 (Stateful) с активной записью

```
[edit]
dmmosharov@dmmosharov-gw-01# run show dhcpv6 server leases
IPv6 address   State   Last communication   Lease expiration   Remaining   Type           Pool               IAID_DUID
-----
2001::198      active  2025/12/06 20:13:53   2025/12/06 22:18:53  2:03:57     non-temporary  dmmosharov-stateful  00:50:01:de:00:01:00:01:30:c7:4b:00:02:42:de:
1:50:00
[edit]
dmmosharov@dmmosharov-gw-01#
```

Рис. 26: Проверка таблицы аренды DHCPv6 (Stateful) с активной записью

Анализ пакетов Stateful DHCPv6 в Wireshark

2 0.003582	fe80::e57:62ff:fe65...	ff02::16	ICMPv6	170 Multicast Listener Report Message v2
3 0.332813	::	ff02::1:ff00:1	ICMPv6	86 Neighbor Solicitation for 2001::1
4 0.595706	fe80::e57:62ff:fe65...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
5 484.617556	fe80::42:dcff:fe52:...	ff02::2	ICMPv6	70 Router Solicitation from 02:42:dc:52:86:00
6 937.575044	fe80::e57:62ff:fe65...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
7 937.585840	fe80::e57:62ff:fe65...	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
8 938.337732	fe80::e57:62ff:fe65...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
9 940.186034	fe80::e57:62ff:fe65...	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
10 957.068438	fe80::e57:62ff:fe65...	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
11 973.089872	fe80::e57:62ff:fe65...	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
12 1488.232834	fe80::42:dcff:fe52:...	ff02::1:2	DHCPv6	118 Solicit XID: 0x779f36 CID: 0001000130c6ff410242dc528600
13 1488.308014	fe80::e57:62ff:fe65...	fe80::42:dcff:fe52:...	DHCPv6	188 Advertise XID: 0x779f36 IAA: 2001::199 CID: 0001000130c6ff410242dc528600
14 1489.284769	fe80::42:dcff:fe52:...	ff02::1:2	DHCPv6	164 Request XID: 0x19720c CID: 0001000130c6ff410242dc528600 IAA: 2001::199
15 1489.349186	fe80::e57:62ff:fe65...	fe80::42:dcff:fe52:...	DHCPv6	188 Reply XID: 0x19720c IAA: 2001::199 CID: 0001000130c6ff410242dc528600
16 1489.451784	fe80::42:dcff:fe52:...	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
17 1489.555528	::	ff02::1:ff00:199	ICMPv6	86 Neighbor Solicitation for 2001::199
18 1489.785466	fe80::42:dcff:fe52:...	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
19 1493.739756	fe80::e57:62ff:fe65...	fe80::42:dcff:fe52:...	ICMPv6	86 Neighbor Solicitation for fe80::42:dcff:fe52:8600 from 0c:57:62:65:00:02
20 1493.740225	fe80::42:dcff:fe52:...	fe80::e57:62ff:fe65...	ICMPv6	78 Neighbor Advertisement fe80::42:dcff:fe52:8600 (sol)
21 1498.889314	fe80::42:dcff:fe52:...	fe80::e57:62ff:fe65...	ICMPv6	86 Neighbor Solicitation for fe80::e57:62ff:fe65:2 from 02:42:dc:52:86:00
22 1498.890765	fe80::e57:62ff:fe65...	fe80::42:dcff:fe52:...	ICMPv6	78 Neighbor Advertisement fe80::e57:62ff:fe65:2 (rtr, sol)
23 1538.201445	fe80::e57:62ff:fe65...	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
24 1606.244819	2001::199	2001::1	ICMPv6	118 Echo (ping) request id=0x0003, seq=1, hop limit=64 (reply in 27)
25 1606.249346	2001::1	ff02::1:ff00:199	ICMPv6	86 Neighbor Solicitation for 2001::199 from 0c:57:62:65:00:02
26 1606.249660	2001::199	2001::1	ICMPv6	86 Neighbor Advertisement 2001::199 (sol, ovr) is at 02:42:dc:52:86:00
27 1606.251028	2001::1	2001::199	ICMPv6	118 Echo (ping) reply id=0x0003, seq=1, hop limit=64 (request in 24)
28 1607.248246	2001::199	2001::1	ICMPv6	118 Echo (ping) request id=0x0003, seq=2, hop limit=64 (reply in 29)
29 1607.252853	2001::1	2001::199	ICMPv6	118 Echo (ping) reply id=0x0003, seq=2, hop limit=64 (request in 28)
30 1611.530679	fe80::42:dcff:fe52:...	2001::1	ICMPv6	86 Neighbor Solicitation for 2001::1 from 02:42:dc:52:86:00
31 1611.530889	fe80::42:dcff:fe52:...	fe80::e57:62ff:fe65...	ICMPv6	86 Neighbor Solicitation for fe80::e57:62ff:fe65:2 from 02:42:dc:52:86:00
32 1611.536343	2001::1	fe80::42:dcff:fe52:...	ICMPv6	78 Neighbor Advertisement 2001::1 (rtr, sol)

Рис. 27: Анализ пакетов Stateful DHCPv6 в Wireshark

В результате выполнения лабораторной работы были получены навыки использования и настройки dhcpv6